



COMPETENCY STANDARD

FOR

**INDUSTRIAL ROBOT OPERATION AND
TROUBLESHOOTING**

(LIGHT ENGINEERING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for **Industrial Robot Operation and Troubleshooting** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as a framework for providing industry-aligned training, ensuring that individuals meeting the prescribed standards through assessments are qualified to secure relevant employment opportunities.

This document has been specifically developed for training purposes under the Skills for Industry Competitiveness and Innovation Program (SICIP), aiming to address the skills requirements of the Light engineering sector. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level 15 & 16)
71-72 Eskaton Garden, Dhaka-1000
Phone: +8802 55138753-55,
Website: www.sicip.gov.bd

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment, which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a glance:

Generic Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-IRO-01-G	Apply Occupational Health and Safety (OHS) Practices in Workplace	1. Identify OHS policies and procedures 2. Practice personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-LE-IRO-02-G	Operate in a Self-Directed Team	1. Identify team goals and work processes 2. Communicate and cooperate with team members. 3. Work as a team member. 4. Solve problems as a team member	10
Total Hour			20

Sector Specific Competencies (10 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-IRO-01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hour			10

Occupation Specific Competencies (270 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-IRO-01-O	Execute Electrical Sequence Circuit and Control using Relay Logic	1. Interpret relay logic operation 2. Perform mechanical switching operation 3. Execute timer and counter application 4. Perform forward reverse and star delta operation	40
SICIP-LE-IRO-02-O	Perform Basic PLC Programming	1. Install PLC and associated Wiring 2. Setup PLC hardware configurations with software tool 3. Create PLC program 4. Design HMI for PLC	45
SICIP-LE-IRO-03-O	Analyze Industrial Sensors	1. Identify types of industrial sensors 2. Perform sensor wiring using correct interface 3. Generate PLC program for data acquisition 4. Perform data analysis on sensor outputs	35
SICIP-LE-IRO-04-O	Perform Electro Pneumatic Sequence Control	1. Understand robot electrical and pneumatic diagrams 2. Design and simulate electropneumatic circuit	55

Code	Unit of Competency	Elements of Competency	Duration (hours)
		3. Carry out electropneumatic circuit connection 4. Generate PLC program for sequence control 5. Operate modular production system (MPS) stations	
SICIP-LE-IRO-05-O	Simulate Robotic Arm in Software	1. Install and configure simulation software 2. Create and configure simulation environment 3. Generate robotic program 4. Transfer program file to robotic arm system	20
SICIP-LE-IRO-06-O	Perform Robotic Program	1. Identify programming syntax 2. Develop program for robot arm movement 3. Develop program for controlling robot input and output 4. Operate robotic arm	50
SICIP-LE-IRO-07-O	Operate Collaborative Robot (COBOT)	1. Understand collaborative robot principles, features and safety protocols 2. Set up and operate collaborative robots 3. Integrate COBOT with CNC machine 4. Install end-effectors with COBOT 5. Program tasks for collaborative robots 6. Troubleshoot and optimize robot performance	25
Total Hours			0

COMPETENCY STANDARD: INDUSTRIAL ROBOT OPERATION AND TROUBLESHOOTING

A. The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-IRO-01-G
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS Policies and Procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice Personal Health and Safety Procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report Hazards and Risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to Emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:

1. OHS Policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations 1.3 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and Risks	3.1 Chemical hazards. 3.2 Biological hazards. 3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.
4. Emergency Response Plans and Procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First Aid Procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices

	2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Using firefighting accessories and fire extinguishers 2.7 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified OHS policies and procedures 1.2 practiced personal health and safety procedures 1.3 reported hazards and risks 1.4 responded to emergencies
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A SELF-DIRECTED TEAM	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-IRO-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Team Goals and Work Processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and Cooperate with Team Members	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a Team Member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practices. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve Problems as a Team Member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range
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	May Include but not limited to:
1. Forms of Communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning. 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Support mechanism to other members of team to ensure achievements of goals. 1.10 Methods of identifying current and potential problems faced by a team 1.11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.5 Using formal and informal forms of communication 2.6 Understanding and valuing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, and workplace procedures and practice. 2.8 Team member's support to other members is appreciated and valued Identifying current and potential problems faced by the team 2.9 Identifying solutions to the problem

	2.10 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Pens 4.3 Papers 4.4 Work books 4.5 Learning manuals

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 communicated and cooperated with team members. 1.3 worked as a team member 1.4 solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

B. The Sector Specific (Common) Competencies

Unit of Competency: APPLY GREEN PRACTICES	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-IRO-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace, and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret Green Concepts	1.1 <u>Principles of green practices</u> are explained. 1.2 <u>Sources of environmental impacts</u> during light engineering activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 <u>Ways of mitigating environmental impacts</u> in light engineering sector are explained.
2. Minimize Resource use in the Workplace	2.1 Water, energy, and raw material consumption are documented. 2.2 <u>Recyclable and non-recyclable items</u> are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement Waste Management Practices	3.1 <u>Different types of wastes</u> are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce wastes in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of Green Practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse

	1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair 1.3 use of sustainable materials with low environmental impact 1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of Environmental Impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution 2.4 Waste generation 2.4.1 waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of Mitigating Environmental Impacts	3.1 Utilizing Energy-Efficient Equipment 3.2 Adopting Renewable Energy Sources 3.3 Implementing Site Protection Measures 3.4 Using Reusable Materials 3.5 Choosing Sustainable Materials 3.6 Using Noise-Reducing Equipment and scheduling work appropriately 3.7 Optimizing Logistics and Delivery
4. Recyclable and Non-Recyclable Items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different Types of Waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Electrical system waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste

	5.8 Chemical West
	5.9 Fuel/Oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in Light Engineering sector 1.2 Key terms and symbols related to environmental sustainability in Light Engineering drawings 1.3 Sources of environmental impacts in Light Engineering 1.4 Methods to minimize resource consumption (water, energy, raw materials) 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying environmental impacts in Light Engineering activities 2.2 Applying methods to minimize resource use in the workplace 2.3 Implementing waste management practices 2.4 Following procedures to safely dispose of hazardous materials 2.5 Documenting water, energy, and material consumption in the workplace 2.6 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools 4.3 Safety gear 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Operation and maintenance manuals 4.8 Waster segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resource use in the workplace 1.3 implemented waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

C. Occupation-specific Competencies

Unit of Competency: EXECUTE ELECTRICAL SEQUENCE CIRCUIT AND CONTROL USING RELAY LOGIC	Nominal Duration: 40 hrs.	Unit Code: SICIP-LE-IRO-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to execute electrical sequence circuit and control using relay logic. It specifically includes the tasks of interpreting relay logic operation, performing mechanical switching operation, executing timer and counter application, and performing forward reverse and star delta operation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret Relay Logic Operation	1.1 Fundamental concepts of relay logic are identified and explained. 1.2 <u>Components of relay logic circuits</u> are identified. 1.3 Relay logic components functions are described. 1.4 Operation of relay logic circuits is interpreted. 1.5 Common relay <u>logic symbols</u> and notations are identified. 1.6 Application of relay logic in industrial automation processes is described.
2. Perform Mechanical Switching Operation	2.1 <u>Mechanical switching devices</u> are identified and described. 2.2 Mechanical switching operation procedures are followed. 2.3 The <u>contact status</u> is checked using multimeter. 2.4 <u>Boolean logic</u> operation using relay is executed. 2.5 Switching devices are operated as per the control system design. 2.6 Mechanical switches are inspected and tested for proper functioning before and after operation.
3. Execute Timer and Counter Application	3.1 <u>Timer</u> and <u>counter</u> are selected based on control system requirements. 3.2 Timer and counter wiring are performed according to electrical diagram. 3.3 Timer and counter settings are made according to application requirements. 3.4 Timer and counter operations are executed following standard procedures.

	3.5 Timer and counter outputs are monitored and verified. 3.6 Faults or deviations in timer and counter performance are identified.
4. Perform Forward Reverse and Star Delta Operation	4.1 Circuit diagrams for latch, DOL, and star-delta configurations are interpreted. 4.2 Latch circuits are constructed by connecting relays, contacts, and coils as per standard wiring diagrams. 4.3 DOL starter circuits are assembled ensuring proper connections for motor start and stop functions. 4.4 Automatic forward and reverse operation are executed using timer and counter. 4.5 Star-delta starter circuits are constructed and star delta operation is executed. 4.6 Faults or abnormalities during operation are identified and reported for corrective actions.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Components of relay logic circuits	1.1 Switches 1.2 Relay 1.3 Magnetic Contactor 1.4 Timer 1.5 Counter 1.6 LED 1.7 Motor 1.8 Sensors 1.9 RCCB 1.10 ELCB 1.11 MCCB
2. Logic symbols	2.1 Normally Open (NO) 2.2 Normally Close (NC) 2.3 Magnetic contactor 2.4 Timer 2.5 Counter 2.6 Motor 2.7 Relay Coil
3. Mechanical switching devices	3.1 Limit switch 3.2 Push button 3.3 Toggle switch 3.4 Selector switch 3.5 Float switch 3.6 Thermal overload relay
4. Contact status	4.1 Normally Open (NO)

	4.2 Normally Close (NC)
5. Boolean logic	5.1 AND 5.2 OR 5.3 NAND 5.4 NOR 5.5 XOR 5.6 XNOR
6. Timer	6.1 On-delay timer 6.2 Off-delay timer
7. Counter	7.1 Up counter 7.2 Down counter

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fundamental Concepts of Relay Logic 1.2 Components of Relay Logic Circuits 1.3 Common Relay Logic Symbols and Notations 1.4 Mechanical Switching Devices 1.5 Boolean Logic Operation Using Relays 1.6 Switching Devices Operation According to Control System Design 1.7 Timer and Counter Settings Based on Application Requirements 1.8 Identifying Faults or Deviations in Timer and Counter Performance 1.9 Latch Circuits Construction 1.10 DOL Starter Circuits Assembly 1.11 Automatic Forward and Reverse Operation 1.12 Star-Delta Starter Circuits Construction and Operation
2. Underpinning Skills	2.1 Following Mechanical Switching Operation Procedures 2.2 Checking Contact Status Using Multimeter 2.3 Executing Boolean Logic Operation Using Relay 2.4 Operating Switching Devices According to Control System Design 2.5 Inspecting and Testing Mechanical Switches for Proper Functioning 2.6 Selecting Timer and Counter Based on Control System Requirements 2.7 Performing Timer and Counter Wiring According to Electrical Diagram 2.8 Setting Timer and Counter Based on Application Requirements 2.9 Executing Timer and Counter Operations Following Standard Procedures 2.10 Monitoring and Verifying Timer and Counter Outputs

	2.11 Executing Automatic Forward and Reverse Operation Using Timer and Counter 2.12 Constructing Star-Delta Starter Circuits and Executing Star-Delta Operation
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities appropriate to the process or activities

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted relay logic operation 1.2 performed mechanical switching operation 1.3 executed timer and counter application 1.4 performed forward reverse and star delta operation
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM BASIC PLC PROGRAMMING	Nominal Duration: 45 Hrs.	Unit Code: SICIP-LE-IRO-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform basic PLC programming. It specifically includes the tasks of Installing PLC and associated wiring, setup PLC hardware configurations with software tool, creating PLC program and designing HMI for PLC.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Install PLC and Associated Wiring	1.1 Total number of inputs and outputs are estimated based on application requirements. 1.2 Suitable <u>PLC type</u> is selected for the application. 1.3 Appropriate <u>input and output devices</u> are chosen for the PLC system. 1.4 Human-Machine Interface (HMI) is selected based on user interface requirements. 1.5 Hardware is configured in the PLC <u>programming software</u> . 1.6 PLC wiring diagram is reviewed and updated. 1.7 Wiring for inputs and outputs is installed and tested according to electrical codes and standards.
2. Setup PLC Hardware Configurations with Software Tool	2.1 PLC software is installed on the designated computer or device. 2.2 Software is licensed and activated according to the software's licensing terms. 2.3 PLC program is downloaded to the PLC hardware from the software tool. 2.4 Configuration settings for hardware communication and <u>I/O modules</u> are defined within the software tool. 2.5 Correct <u>communication protocols</u> are selected and configured. 2.6 System's firmware is updated, if necessary.
3. Create PLC program	3.1 New program file is created within the PLC programming software. 3.2 Input and output tags are properly assigned in the program. 3.3 Ladder diagram is constructed for digital inputs and outputs.

	3.4 Program is verified using simulation tools as expected before implementation. 3.5 Program is optimized for efficiency and resource usage.
4. Design HMI for PLC	4.1 HMI is configured with the correct communication settings. 4.2 Communication between the HMI and PLC is established and tested. 4.3 HMI interface is designed according to requirement. 4.4 HMI tag addresses are assigned to match the PLC's input and output tags. 4.5 HMI is operated to control digital inputs and outputs, verifying that the system responds.

Range of Variables

Variable	Range (Includes but not limited to:)
1. PLC type	1.1 Compact 1.2 Modular
2. Input and output devices	2.1 Input modules 2.2 Output modules 2.3 Communication modules 2.4 Switch 2.5 Sensor 2.6 Light 2.7 Buzzer 2.8 Potentiometer 2.9 Valve
3. Programming software	3.1 TIA portal 3.2 FluidSim 3.3 GXworks 3.4 Step 7 3.5 WinCC
4. I/O modules	4.1 Digital input module 4.2 Digital output module 4.3 Analog input module 4.4 Analog output module 4.5 RTD module
5. Communication protocols	5.1 Ethernet 5.2 USB 5.3 RS232 5.4 RS485

Curricular Content Guide

1. Underpinning Knowledge	1.1 Selection of Suitable PLC Type 1.2 Choosing Appropriate Input and Output Devices 1.3 Selection of Human-Machine Interface (HMI)
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	1.4 Hardware Configuration in PLC Programming Software 1.5 Definition of Configuration Settings for Hardware Communication and I/O Modules 1.6 Creation of New Program File in PLC Programming Software 1.7 Construction of Ladder Diagram for Digital Inputs and Outputs 1.8 Verification of Program Using Simulation Tools 1.9 Optimization of Program for Efficiency and Resource Usage 1.10 Establishment and Testing of Communication Between HMI and PLC 1.11 Design of HMI Interface According to Requirements 1.12 Operation of HMI to Control Digital Inputs and Outputs
2. Underpinning Skills	2.1 Estimating Total Number of Inputs and Outputs Based on Application Requirements 2.2 Choosing Appropriate Input and Output Devices for the PLC System 2.3 Selecting Human-Machine Interface (HMI) Based on User Interface Requirements 2.4 Configuring Hardware in the PLC Programming Software 2.5 Reviewing and Updating the PLC Wiring Diagram 2.6 Installing and Testing Wiring for Inputs and Outputs According to Electrical Codes and Standards 2.7 Installing PLC Software on the Designated Computer or Device 2.8 Selecting and Configuring Correct Communication Protocols 2.9 Creating a New Program File in the PLC Programming Software 2.10 Operating the HMI to Control Digital Inputs and Outputs, Verifying System Response
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual)

	4.2 Tools, equipment, materials and facilities appropriate to the process or activities
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 installed PLC and associated wiring 1.2 setup PLC hardware configurations with software tool 1.3 created PLC program 1.4 designed HMI for PLC
2 Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3 Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: ANALYZE INDUSTRIAL SENSORS	Nominal Duration: 35 hrs.	Unit Code: SICIP-LE-IRO-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to analyze industrial sensors. It specifically includes the tasks of identifying types of industrial sensors, performing sensor wiring using correct interface, generating PLC program for data acquisition and performing data analysis on sensor outputs.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Types of Industrial Sensors	1.1 Types of <u>industrial sensors</u> are interpreted. 1.2 Classifications of industrial sensors are identified and described. 1.3 Function and role of sensors are explained in relation to industrial applications. 1.4 Applications of industrial sensors are interpreted.
2. Perform Sensor Wiring using Correct Interface	2.1 Appropriate sensor is selected based on the specific application requirements. 2.2 <u>Sensor's communication protocols</u> are identified and correctly understood. 2.3 Sensor is wired using the <u>wiring interface</u> according to the sensor's specifications. 2.4 Wiring diagram is followed for signal integrity and safety. 2.5 Sensor's wiring is checked for proper polarity, grounding, and shielding. 2.6 Sensor wiring is tested and verified to the control system.
3. Generate PLC Program for Data Acquisition	3.1 Sensor pinout is identified by referring to the datasheet. 3.2 PLC program is written to read or write sensor data. 3.3 Sensor calibration is performed according to data sheet. 3.4 Raw data is converted ensuring that the data is scaled and formatted.
4. Perform Data Analysis on Sensor Outputs	4.1 Sensor data is recorded and logged, ensuring accurate capture of sensor readings for analysis. 4.2 Sensor readings are compared for accuracy and verification. 4.3 Signals from sensor data are processed, ensuring proper filtering, scaling, and conversion for analysis. 4.4 Sensor data is displayed on the HMI to usable format.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Industrial Sensors	1.1 Optical 1.2 Limit Switch 1.3 Pressure 1.4 Temperature 1.5 Level 1.6 Flowrate 1.7 Humidity 1.8 Capacitive 1.9 Inductive
2. Sensor's Communication Protocols	2.1 Modbus RTU/TCP/IP 2.2 Profibus 2.3 Ethernet 2.4 IO link 2.5 Highway Addressable Remote Transducer (HART) 2.6 Universal Asynchronous Receiver/Transmitter/Transistor-Transistor Logic (UART/TTL)
3. Wiring Interface	3.1 PNP (sourcing) 3.2 NPN (sinking) 3.3 Push-Pull 3.4 4-20 mA loop 3.5 0-10 volt 3.6 RS-485 3.7 RTD (Resistance Temperature Detector)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Interpretation of Types of Industrial Sensors 1.2 Identification and Description of Classifications of Industrial Sensors 1.3 Explanation of Function and Role of Sensors in Industrial Applications 1.4 Interpretation of Applications of Industrial Sensors 1.5 Selection of Appropriate Sensor Based on Application Requirements 1.6 Identification and Understanding of Sensor Communication Protocols 1.7 Wiring of Sensor Using the Wiring Interface According to Specifications 1.8 Comparison of Sensor Readings for Accuracy and Verification 1.9 Processing Signals from Sensor Data for Proper Filtering, Scaling, and Conversion
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2. Underpinning Skills	2.1 Selecting the Appropriate Sensor Based on Specific Application Requirements 2.2 Wiring the Sensor Using the Interface According to Specifications 2.3 Following the Wiring Diagram for Signal Integrity and Safety 2.4 Checking Sensor's Wiring for Proper Polarity, Grounding, and Shielding 2.5 Testing and Verifying Sensor Wiring to the Control System 2.6 Writing PLC Program to Read or Write Sensor Data 2.7 Performing Sensor Calibration According to Datasheet 2.8 Converting Raw Data with Proper Scaling and Formatting 2.9 Recording and Logging Sensor Data for Accurate Capture 2.10 Comparing Sensor Readings for Accuracy and Verification 2.11 Processing Sensor Signals for Filtering, Scaling, and Conversion 2.12 Displaying Sensor Data on the HMI in Usable Format
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4 Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools and equipment required for the job 4.3 Materials required 4.4 Instructional manuals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified types of industrial sensors 1.2 performed sensor wiring using correct interface 1.3 generated PLC program for data acquisition 1.4 performed data analysis on sensor outputs
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical demonstration

	2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ELECTRO PNEUMATIC SEQUENCE CONTROL	Nominal Duration: 55 hrs.	Unit Code: SICIP-LE-IRO-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform electro pneumatic sequence control. It specifically includes the tasks of understanding robot electrical and pneumatic diagrams, designing and simulate electropneumatic circuit, carry out electropneumatic circuit connection, generating PLC program for sequence control and operating modular production system (MPS) stations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Understand Robot Electrical and Pneumatic Diagrams	1.1 Safety circuits in robotics are recognized to protect operators and equipment. 1.2 <u>Basic electrical components in robots</u> are identified. 1.3 <u>Circuit diagrams</u> are read and interpreted. 1.4 Control systems in robots are understood of robots through electrical and pneumatic control systems.
2. Design and Simulate Electropneumatic Circuit	2.1 <u>Electropneumatic components</u> are selected. 2.2 <u>Circuit design software</u> is installed for designing and simulating the circuit. 2.3 Circuits are designed in the software according to operational requirements. 2.4 Circuits are simulated in the software, ensuring functionality is tested and validated.
3. Carry out Electropneumatic Circuit Connection	3.1 Correct pressure settings are applied as per operational manual. 3.2 Electric wiring is performed according to designed circuit. 3.3 <u>Input and output devices</u> are connected in the electropneumatic system. 3.4 Pneumatic components are installed properly. 3.5 <u>Pneumatic cylinders</u> are operated in sequence according to the system design.
4. Generate PLC Program for Sequence Control	4.1 Ladder logic is developed ensuring logical flow for sequence control. 4.2 PLC is programmed for sequence control. 4.3 HMI is designed for sequential operation. 4.4 Emergency stop button is programmed ensuring that pressing the button halts the process.
5. Operate Modular Production System	5.1 <u>MPS stations</u> are commissioned following the manufacturer's manual and standard safety procedures.

(MPS) Stations	5.2 PLC for the MPS station is programmed based on the required process logic or task specification. 5.3 Sensors are tested, calibrated and verified to ensure accurate and reliable detection of work pieces. 5.4 <u>MPS station modules</u> are operated in the correct functional sequence as per the defined process workflow. 5.5 An HMI (Human-Machine Interface) is designed and configured to enable user-friendly monitoring and control of MPS station operations.
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Range of Variables

Variable	Range (Includes but not limited to:)
1. Basic Electrical Components in Robots	1.1 Supply module 1.2 Control unit 1.3 Signal distributor 1.4 Operator panel 1.5 Teaching box
2. Circuit Diagrams	2.1 Input/output wiring diagram 2.2 Safety circuit diagram 2.3 Power supply diagram 2.4 Pneumatic control diagram
3. Electropneumatic Components	3.1 Filter-Regulator-Lubricator (FRL) units 3.2 Pressure Gauges 3.3 Solenoid Valves 3.4 Switches 3.5 Sensors
4. Circuit Design Software	4.1 FluidSIM 4.2 Automation studio
5. Input and Output Devices	5.1 Optical sensors 5.2 Inductive proximity sensors 5.3 Capacitive proximity sensors 5.4 Push button 5.5 Selector switch 5.6 Solenoid relay 5.7 DC motor
6. Pneumatic Cylinders	6.1 Single acting cylinder 6.2 Double acting cylinder
7. MPS Stations	7.1 Distribution stations 7.2 Sorting stations 7.3 Robot stations
8. MPS Station Modules	8.1 Conveyor module 8.2 Detection module 8.3 Robot assembly module 8.4 Stacking magazine module

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identification of Basic Electrical Components in Robots 1.2 Reading and Interpretation of Circuit Diagrams 1.3 Understanding Control Systems in Robots (Electrical and Pneumatic) 1.4 Selection of Electropneumatic Components 1.5 Installation of Circuit Design Software for Simulation 1.6 Operation of Pneumatic Cylinders in Sequence According to Design 1.7 Design of HMI for Sequential Operation 1.8 PLC Programming for MPS Station Based on Process Logic or Task Specification 1.9 Operation of MPS Station Modules in Correct Functional Sequence
2. Underpinning Skills	2.1 Selecting Electropneumatic Components 2.2 Installing Circuit Design Software for Designing and Simulating Circuits 2.3 Simulating Circuits in the Software to Test and Validate Functionality 2.4 Applying Correct Pressure Settings as per Operational Manual 2.5 Performing Electric Wiring According to Designed Circuit 2.6 Connecting Input and Output Devices in the Electropneumatic System 2.7 Operating Pneumatic Cylinders in Sequence According to System Design 2.8 Programming Emergency Stop Button to Halt Process 2.9 Programming PLC for MPS Station Based on Process Logic or Task Specification 2.10 Testing, Calibrating, and Verifying Sensors for Accurate and Reliable Detection 2.11 Designing and Configuring HMI for User-Friendly Monitoring and Control of MPS Operations
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities needed for the process/ activity 4.3 Materials required 4.4 Instructional manual
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 understood robot electrical and pneumatic diagrams 1.2 designed and simulated electropneumatic circuit 1.3 carried out electropneumatic circuit connection 1.4 generated PLC program for sequence control 1.5 operated modular production system (MPS) stations
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: SIMULATE ROBOTIC ARM IN SOFTWARE	Nominal Duration: 20 hrs.	Unit Code: SICIP-LE-IRO-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to simulate robotic arm in software. It specifically includes the tasks of installing and configuring simulation software, creating and configuring simulation environment, generating robotic program and transferring program file to robotic arm system.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Install and Configure Simulation Software	1.1 <u>Robot simulation software</u> is downloaded for the system's specifications. 1.2 Robot simulation software is installed and all components are set up. 1.3 Software licensing is performed for functionality and compliance. 1.4 System configuration is completed with the required hardware and settings.
2. Create and Configure Simulation Environment	2.1 3D simulation environment is created using built-in objects or imported models. 2.2 Software settings are configured for robot simulation. 2.3 Appropriate robot model is selected for the task. 2.4 External 3D models are imported into the simulation. 2.5 Custom 3D object is created to simulate pick-and-place tasks. 2.6 Sensors and actuators are added in the environment. 2.7 Pick and place target positions are defined and assigned in the simulation environment.
3. Generate Robotic Program	3.1 Robot end effectors are assigned and linked to the robot model. 3.2 Robot I/O (input/output) mapping is configured for simulation control. 3.3 Robot motion and logic are programmed using available <u>tools</u> . 3.4 Simulation is executed to test and visualize robot behavior. 3.5 Programming languages supported by simulation software are identified. 3.6 Program is generated/exported into supported format. 3.7 Program is reviewed and edited for corrections or optimization.

4. Transfer Program File to Robotic Arm System	4.1	Simulation software is integrated with PLC programming software.
	4.2	Supported programming languages are identified with the robotic arm's control system.
	4.3	Program is generated into the supported format for the robotic arm's controller.
	4.4	Program is edited for any corrections and resolved before transfer.
	4.5	Program is transferred into the robot controller.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Robot simulation software	1.1 CIROS studio
	1.2 MELSOFT RT Toolbox 3
2. Tools	2.1 RT Toolbox
	2.2 Scripting

Curricular Content Guide

1. Underpinning Knowledge	1.1	Installation and Setup of Robot Simulation Software
	1.2	Software Licensing for Functionality and Compliance
	1.3	Creation of 3D Simulation Environment Using Built-in Objects or Imported Models
	1.4	Selection of Appropriate Robot Model for the Task
	1.5	Importation of External 3D Models into the Simulation
	1.6	Addition of Sensors and Actuators in the Simulation Environment
	1.7	Definition and Assignment of Pick-and-Place Target Positions
	1.8	Assignment and Linking of Robot End Effectors to the Robot Model
	1.9	Identification of Programming Languages Supported by Simulation Software
	1.10	Reviewing and Editing Program for Corrections or Optimization
	1.11	Identification of Supported Programming Languages with Robotic Arm's Control System
	1.12	Editing Program for Corrections Before Transfer
2. Underpinning Skills	2.1	Downloading Robot Simulation Software According to System Specifications
	2.2	Installing Robot Simulation Software and Setting Up Components
	2.3	Performing Software Licensing for Functionality and Compliance
	2.4	Completing System Configuration with Required

	Hardware and Settings 2.5 Creating a 3D Simulation Environment Using Built-In Objects or Imported Models 2.6 Importing External 3D Models into the Simulation 2.7 Creating Custom 3D Objects to Simulate Pick-and-Place Tasks 2.8 Configuring Robot I/O Mapping for Simulation Control 2.9 Programming Robot Motion and Logic Using Available Tools 2.10 Executing the Simulation to Test and Visualize Robot Behavior 2.11 Generating and Exporting Program into a Supported Format 2.12 Integrating Simulation Software with PLC Programming Software
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities needed for the process/activity 4.3 Materials relevant to the proposed activity

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 installed and configure simulation software 1.2 created and configure simulation environment 1.3 generated robotic program 1.4 transferred program file to robotic arm system
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: PERFORM ROBOTIC PROGRAM	Nominal Duration: 50 hrs.	Unit Code: SICIP-LE-IRO-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform robotic program. It specifically includes the tasks of identifying programming syntax, developing program for robot arm movement, developing program for controlling robot input and output and operating robotic arm.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Identify Programming Syntax	1.1 Correct syntax for defining variables is identified. 1.2 Required command structures for programming the robotic arm are recognized. 1.3 Proper use of <u>operators</u> and logical conditions is identified. 1.4 Structure for <u>Continuous</u> and <u>conditional statements</u> is defined.
2. Develop Program for Robot Arm Movement	2.1 Robot's starting position is defined in the program. 2.2 Appropriate joint angles or coordinates are set for arm movement. 2.3 Motion paths for linear or rotational movements are programmed. 2.4 Speed and acceleration parameters are specified for each movement. 2.5 End effectors are programmed using teach pendant.
3. Develop Program for Controlling Robot Input and Output	3.1 Input devices are configured to trigger robot actions. 3.2 Input data is read and decisions are made based on defined conditions. 3.3 Output devices are activated based on logic conditions. 3.4 Communication protocols for input/output devices are established and connected. 3.5 Robot programming is performed for <u>specific tasks</u> . 3.6 Error codes from the robot's controller are read and analyzed. 3.7 Debugging tools are used to monitor and fix the logic flow.
4. Operate Robotic Arm	4.1 Safety guidelines are followed as per safety manual. 4.2 <u>Robot peripherals</u> are installed and connected. 4.3 Robot Tool Center Point (TCP) is configured correctly as per instruction manual. 4.4 Robot jogging is performed using teach pendant.

	4.5 Robot axes are aligned using teach pendant.
	4.6 Pre-programmed positions are taught using teach pendant.
	4.7 Robot program is selected and executed for operation.

Range of Variables

Variable	Range (Includes but not limited to:)
1. Operators	1.1 Arithmetic 1.2 Relational 1.3 Logical
2. Continuous	2.1 Loop 2.2 For 2.3 While
3. Conditional statements	3.1 If elseif 3.2 Case
4. Specific tasks	4.1 Pick and place 4.2 Palletizing 4.3 Color sorting 4.4 Screwing 4.5 Palletizing 4.6 Conveyor tracking
5. Robot peripherals	5.1 End effector 5.1.1 Vacuum 5.1.2 Electrical 5.2 Sensor 5.2.1 Proximity 5.2.2 Photo electric 5.2.3 Force torque 5.3 Safety curtain/fence/door

Curricular Content Guide

1. Underpinning Knowledge	1.1 Identification of Correct Syntax for Defining Variables 1.2 Recognition of Required Command Structures for Robotic Arm Programming 1.3 Identification of Proper Use of Operators and Logical Conditions 1.4 Definition of Structure for Continuous and Conditional Statements 1.5 Definition of Robot's Starting Position in the Program 1.6 Programming End Effectors Using Teach Pendant 1.7 Configuration of Input Devices to Trigger Robot Actions 1.8 Reading Input Data and Making Decisions Based on Defined Conditions 1.9 Activation of Output Devices Based on Logic Conditions
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	1.10 Reading Error Codes from Robot Controller 1.11 Following Safety Guidelines as per Safety Manual
2. Underpinning Skills	2.1 Setting Appropriate Joint Angles or Coordinates for Arm Movement 2.2 Programming Motion Paths for Linear or Rotational Movements 2.3 Specifying Speed and Acceleration Parameters for Each Movement 2.4 Programming End Effectors Using Teach Pendant 2.5 Configuring Input Devices to Trigger Robot Actions 2.6 Activating Output Devices Based on Logic Conditions 2.7 Establishing and Connecting Communication Protocols for I/O Devices 2.8 Performing Robot Programming for Specific Tasks 2.9 Reading and Analyzing Error Codes from the Robot's Controller 2.10 Following Safety Guidelines as per the Safety Manual 2.11 Installing and Connecting Robot Peripherals 2.12 Configuring Robot Tool Center Point (TCP) as per Instruction Manual 2.13 Performing Robot Jogging Using Teach Pendant
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities needed for the process/activity 4.3 Materials relevant to the proposed activity

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified programming syntax 1.2 developed program for robot arm movement 1.3 developed program for controlling robot input and output
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	1.4 operated robotic arm
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE COLLABORATIVE ROBOT (COBOT)	Nominal Duration: 25 hrs.	Unit Code: SICIP-LE-IRO-07-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate collaborative robot (COBOT). It specifically includes the tasks of understanding collaborative robot principles, features and safety protocols, setting up and operating collaborative robots, integrating COBOT with CNC machine, installing end-effectors with COBOT, programing tasks for collaborative robots and troubleshooting and optimizing robot performance.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are described in the range of variables).

Elements of Competency	Performance Criteria
1. Understand Collaborative Robot Principles, Features and Safety Protocols	1.1 Collaborative robot principles and applications are explained. 1.2 <u>COBOT features</u> are described and interpreted as per manufacturer specifications. 1.3 Safety considerations for collaborative robots are explained. 1.4 Safety regulations and risk assessment procedures are identified with workplace safety protocols.
2. Set up and Operate Collaborative Robots	2.1 COBOT is assembled, mounted and powered as per operation manual. 2.2 Robot control panel and teach pendant interface are understood, ensuring familiarity with system controls. 2.3 Calibration and basic configuration of the COBOT are performed as per manual. 2.4 <u>End-effectors</u> are installed and configured with the COBOT, ensuring the correct attachment and setup for the required operation.
3. Integrate COBOT with CNC Machine	3.1 CNC input/output terminals are identified for <u>specific operation</u> as per manual. 3.2 Signal wiring is performed through the CNC machine to the COBOT for communication between the two systems. 3.3 Input/output signals are configured using the COBOT teach pendant ensuring proper signal mapping.
4. Program Tasks for Collaborative Robots	4.1 Motion programs are created and executed using the teach pendant. 4.2 Tool center points (TCP) and payload are configured according to tasks requirement. 4.3 Collaborative tasks are programmed using teach pendant.
5. Troubleshoot and	5.1 <u>Common issues</u> with collaborative robot operation are

Optimize Robot Performance	identified. 5.2 <u>Robot parameters</u> are optimized 5.3 <u>Maintenance</u> and inspection for COBOT are performed as per manual.
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Range of Variables

Variable	Range (Includes but not limited to:)
1. COBOT features	1.1 Payload 1.2 Range 1.3 Joint speed 1.4 Accuracy and Repeatability 1.5 Reach
2. End-effectors	2.1 Gripper 2.2 Vacuum cup 2.3 Welding torch 2.4 Screwdriver spindle
3. Specific operation	3.1 Door opening 3.2 Door closing 3.3 Cycle starting 3.4 Blowing air 3.5 Job pick and place 3.6 Job palletizing 3.7 Vise opening and closing
4. Common issues	4.1 Misalignment 4.2 Sensor errors 4.3 Software malfunctions
5. Robot parameters	5.1 Robot speed 5.2 Precision 5.3 Cycle time
6. Maintenance	6.1 Schedule maintenance 6.2 Preventive maintenance 6.3 Breakdown maintenance

Curricular Content Guide

1. Underpinning Knowledge	1.1 Explanation of Collaborative Robot Principles and Applications 1.2 Description and Interpretation of COBOT Features as per Manufacturer Specifications 1.3 Explanation of Safety Considerations for Collaborative Robots 1.4 Identification of Safety Regulations and Risk Assessment Procedures 1.5 Understanding of Robot Control Panel and Teach Pendant Interface
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	1.6 Installation and Configuration of End Effectors with COBOT 1.7 Identification of CNC Input/Output Terminals for Specific Operation 1.8 Creation and Execution of Motion Programs Using Teach Pendant 1.9 Programming Collaborative Tasks Using Teach Pendant 1.10 Identification of Common Issues with Collaborative Robot Operation
2. Underpinning Skills	2.1 Assembling, Mounting, and Powering the COBOT as per the Operation Manual 2.2 Performing Calibration and Basic Configuration of the COBOT as per the Manual 2.3 Installing and Configuring End-Effectors with the COBOT for Correct Attachment and Setup 2.4 Performing Signal Wiring through the CNC Machine to the COBOT for Communication 2.5 Configuring Input/Output Signals Using the COBOT Teach Pendant for Proper Signal Mapping 2.6 Configuring Tool Center Points (TCP) and Payload According to Task Requirements 2.7 Optimizing Robot Parameters 2.8 Performing Maintenance and Inspection for COBOT as per the Manual
3. Underpinning Attitudes	1.1 Commitment to occupational health and safety 1.2 Promptness in carrying out activities 1.3 Sincerity and honesty to duties 1.4 Environmental concerns 1.5 Eagerness to learn 1.6 Tidiness and timeliness 1.7 Respect for rights of peers and seniors in workplace 1.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities needed for the process/activity 4.3 Materials relevant to the proposed activity

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 understood collaborative robot principles, features and safety protocols 1.2 set up and operated collaborative robots 1.3 integrated COBOT with CNC machine 1.4 installed end-effectors with COBOT 1.5 programed tasks for collaborative robots 1.6 troubleshooted and optimized robot performance
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Workshop/Lab Facility Standard

Course Name:	Industrial Robot Operation and Troubleshooting
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

- Classroom – 350 sft
- Workshop/ lab – 1000 sft

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptop with internet facility	01
2.	Multimedia projector & screen	01
3.	Printer	01
4.	Computer/laptop with licensed software (FluidSIM, Ciros Studio, TIA portal, Automation Studio)	25
5.	COBOT	01
6.	Industrial robot	04
7.	Distribution station	04
8.	Sorting station	04
9.	Air compressor	05
10.	CNC machine	01
11.	End effectors	02
12.	Electro pneumatic training system	10
13.	PLC and sensor training system	05
14.	Relay Modules	25
15.	Wiring Kits	25
16.	Measurement Devices (Multimeters. Clamp ammeters, Tester etc)	25
17.	HMI Module	25
18.	Digital I/O modules	25
19.	Industrial Sensor (proximity, temperature, pressure)	25

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Industrial Robot Operation and Troubleshooting, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.